Axioms in Voting Theory

and Kenneth May's life and theorem

Tim Hua

Slides avaliable on timhua.me/slides.pdf

December 2021

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2. Notation and Definitions
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  - For example, I will introduce the mathematically defined axiom that represents "treat all voters equally."
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- They're fun and neat mathematically.

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Notation

- N is a finite set of n voters.
- A is a finite set of m alternatives (e.g. candidates in an election).
- Each voter i casts a ballot that lists their complete, transitive, and antireflexive preferences $\succ_i$ over A. We denote the $m!$ possible orderings by $\mathcal{L}(A)$.

The source is Moulin 2016 unless otherwise stated

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- We can always discard some of this information when considering a voting rule (e.g., majority voting only considers the first choice)
- A voting profile $P = (\succ_1, \succ_2, \ldots \succ_n)$ is the set of all ballots by all voters. We denote the set of all possible ballots by $\mathcal{L}(A)^n$.

Definition

Definition

A social choice function, voting rule, or SCF, is a function $f : \mathcal{L}(A)^n \to 2^A$ that returns some subset of A, the alternatives. If $|f(P)| = 1$, we say that f is single-valued and write $f(P) = a$.

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A social welfare function $g : \mathcal{L}(A)^n \to \mathcal{R}(A)$ returns a ranking over the alternatives.

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Different versions of Unanimity

Definition

An SCF f is *unanimous* if, whenever there exists an alternative a such that for all voters i and all alternatives $b \in A \setminus \{a\}$, $a \succ_i b$, then $f(P) = a$.

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An SCF f is *unanimous* if, whenever there exists an alternative a such that for all voters i and all alternatives $b \in A \setminus \{a\}$, $a \succ_i b$, then $f(P) = a$.

In other words, if a is the top choice for every voter, then the social choice function selects it. Before introducing the next axiom, we first define what we mean by Pareto dominated: An alternative x is Pareto dominated if there exists some y such that $y \succ_i x$ for all voters i.

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Definition

An SCF f is *Pareto* if $f(P)$ never contains a Pareto dominated alternative.

Anonymity

Definition

An SCF f is *anonymous* if each voter plays the same role. That is, for a given P, if P' is obtained by swapping i and j's ballots (i.e. $\succ_i' = \succ_j$ and $\succ_j' = \succ_i$), then $f(P) = f(P')$. A voting rule is *dictatorial* if $f(P)$ is the top choice of some voter i.

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Lemma

The U.S. presidential election is not anonymous.

Profiles Under Anonymity

If every voter is identical, then we can express their preferences using these profile tables:

Number of voters	10	8	3	1
First place	a	b	d	c
Second place	b	a	b	d
Third place	c	c	a	b
Fourth place	d	d	c	a

Neutrality and Nonimposition

Definition

An SCF f is *neutral* if alternatives are interchangeable. That is, if $f(P) = a_1$, and P' is obtained by swapping a_1 and a_j in every voters' preferences, then $f(P') = a_j$. f is *imposed* if there exists some $a \in A$ such that $f(P) \neq a$ for all $P \in \mathcal{L}(A)^n$.

Monotonicity

There are many different ways to think about Monotonicity. The general idea is that if the winning candidate gains more/do not lose any support as we move from profile P to P', that candidate should still win.

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Definition (Monotonicity/ Positive Responsiveness)

If $a \in f(P)$, and when some voter n for whom a is not their top choice changes their preferences such that a is their top choice, then under the new preference P', $f(P') = a$

Unrestricted Domain/ Resolute

Definition

We say that an SCF f has the unrestricted domain quality or is resolute if it is single valued for all $P \in \mathcal{L}(A)^n$.

Strategyproofness

Definition

When a voter i changes their preferences from $\succeq_i$ to $\succeq_i'$, and the resulting profile changes from P to P' as a result, we say an SCF f is single-voter strategy-proof if $f(P) \succeq_i f(P')$.

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In this case, $\succeq_i$ is thought of as i's sincere preferences. When i changes their preferences to $\succeq_i'$, the voting rule should not pick a candidate that results in a better outcome under the original, sincere preferences by acting on misrepresented information.

SHAKE SHACK
Tasty Burger

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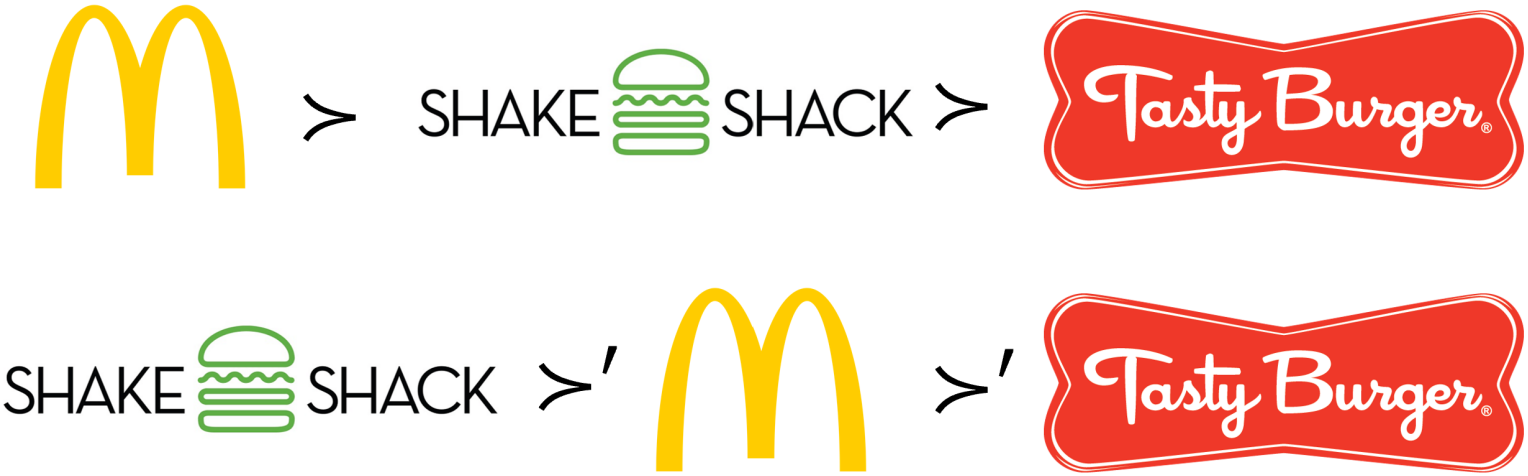
Tasty Burger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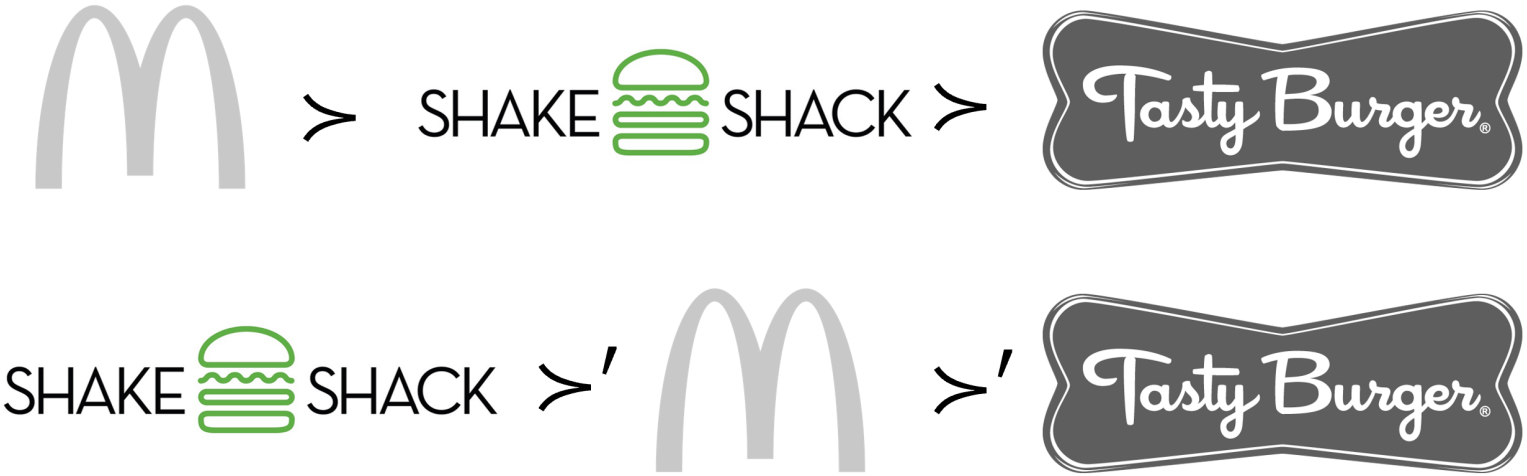
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Summary

- Anonymity: Voters are equal
- Neutrality: Candidates are equal
- Monotonicity: More support = good
- Resolute: Always picks a winner
- Strategyproofness: Can't lie and get a better outcome

So many axioms!

System	Mono-tonic	Condorcet winner	Majo-rity	Condorcet loser	Majority loser	Mutual majority	Smith	ISDA	LIIA	Independence of clones	Reversal symmetry	Participation, consistency	Later-no-harm	Later-no-help	Polynomial time	Resol-vability
Schulze	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	No	No	Yes	Yes
Ranked pairs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	Yes
Split Cycle	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	Yes	No	No	No	Yes	No
Tideman's Alternative	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes	Yes
Kemeny–Young	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes
Copeland	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	No	Yes	No	No	No	Yes	No
Nanson	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	Yes	No	No	No	Yes	Yes
Black	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	Yes	No	No	No	Yes	Yes
Instant-runoff voting	No	No	Yes	Yes	Yes	Yes	No	No	No	Yes	No	No	Yes	Yes	Yes	Yes
Smith/IRV	No	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No	Yes	No	No	No	No	Yes	Yes
Borda	Yes	No	No	Yes	Yes	No	No	No	No	No	Yes	Yes	No	Yes	Yes	Yes
Geller-IRV	No	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	Yes	Yes
Baldwin	No	Yes	Yes	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	Yes	Yes
Bucklin	Yes	No	Yes	No	Yes	Yes	No	No	No	No	No	No	No	Yes	Yes	Yes
Plurality	Yes	No	Yes	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes	Yes
Contingent voting	No	No	Yes	Yes	Yes	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Coombs[3]	No	No	Yes	Yes	Yes	Yes	No	No	No	No	No	No	No	No	Yes	Yes
MiniMax	Yes	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	Yes	Yes
Anti-plurality[3]	Yes	No	No	No	Yes	No	No	No	No	No	No	Yes	No	No	Yes	Yes
Sri Lankan contingent voting	No	No	Yes	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Supplementary voting	No	No	Yes	No	No	No	No	No	No	No	No	No	Yes	Yes	Yes	Yes
Dodgson[3]	No	Yes	Yes	No	No	No	No	No	No	No	No	No	No	No	No	Yes

Figure: (Wikipedia contributors 2021)

Two quick theorems

Theorem (Campbell-Kelly)

If there exists a Condorcet winner—some candidate that beats every other candidate in all pairwise contests—then picking that winner is the unique resolute, anonymous, neutral, and strategyproof SCF (given three or more candidates and an odd number of voters)

Theorem (Gibbard-Satterthwaite)

Any resolute, nonimposed, and strategyproof SCF for three or more alternatives must be a dictatorship.

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In the two candidate case, a social choice function is resolute, neutral, anonymous, and positively responsive if and only if it is simple majority rule.

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Proof.

It's easy to see that simple majority rule is resolute, neutral, anonymous, and positively responsive (again, ignoring ties for now).
We see that anonymity implies that the SCF is dependent only on total number of votes for either candidate. If a voting rule is not simple majority, then there must be some case where a gets more votes than b, but b wins (e.g., a gets seven votes and b gets six). Now we add votes to b until it has exactly as many votes as a does in the beginning (e.g., following the previous example, that means a get six votes and b gets seven). □

May's Theorem Proof continued

Proof (continued).

In this case, positive responsiveness states that b still wins after receiving more votes. However, neutrality states that a must win, since this is the exact same condition as before, but a and b swapped places. We see that a non-simple majority rule SCF cannot satisfy all four axioms. □

Source : (K. O. May 1952)

Reflections on May's Theorem

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- It verifies that simple majority rule makes sense in the axiomatic framework.
- It provides an argument for why we may want to reduce a multicandidate election down to pairwise contests.

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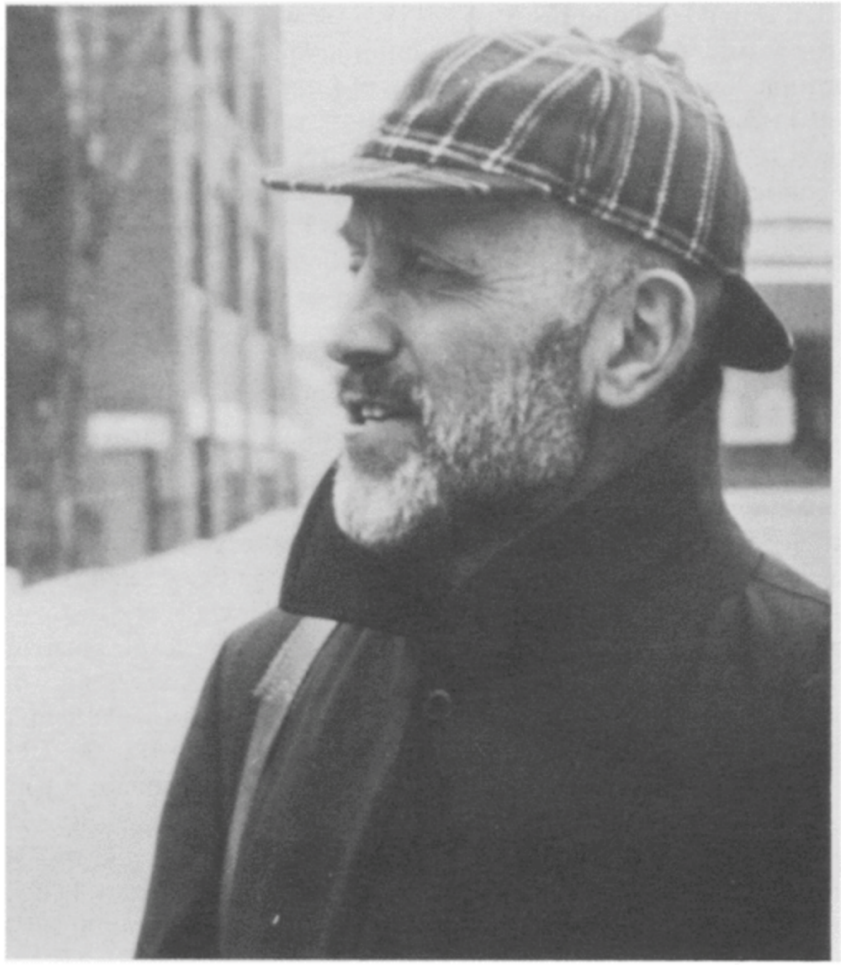
Kenneth O May (1915-1977)



JULY 8, 1915–DECEMBER 1, 1977

Source: (Tropp 1979)

Kenneth O May (1915-1977)

- His dad fought in WWI, came back and got a master's from Columbia, and taught at Dartmouth and then UC Berkeley in the Political Science department. He taught Earl Warren, who later became Chief Justice.

Source on May's life unless otherwise stated is (Jones, Enros, and Tropp 1984)

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- May was a mathematics major at U.C. Berkeley and "had an interest in the mathematical basis of statistics and the use of statistics as an element in national planning"
- Yep. Kenneth is a communist and joined the communist party in college.
- He also pushed for racial integration at a barbershop that refused to serve Black students.

Kenneth O May's Life, continued

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- He got married in London in 1938. His dad tried to stop him and his fellowship forbids fellows from marrying, but he did it anyway.
- So he lost his fellowship and went to Paris, where he studied at the University of Sorbonne and took classes at the l'Université Ouvrière (Worker's University) at night.

l'Université Ouvrière

4. PLANISME ET SOCIALISME

Le développement du système de production capitaliste engendre des crises économiques périodiques. Depuis 1825 éclate, environ tous les dix ans, une de ces crises dont les manifestations ont été résumées en ces termes par Engels :

> *Le commerce s'arrête ; les marchés s'encombrent, les produits sont là, aussi abondants qu'invendables ; la monnaie se cache ; le crédit s'évanouit ; les fabriques se ferment ; les masses ouvrières manquent de moyens d'existence ; les faillites succèdent aux faillites, les ventes forcées aux ventes forcées* [1].

En 1929, le monde capitaliste est entré dans une crise économique sans précédent aussi bien par sa longueur, par sa profondeur, que par l'étendue des dévastations qu'elle a provoquées.

Si bien que le problème des crises constitue un des problèmes essentiels de la politique contemporaine. Comment mettre fin aux crises? Que faire pour qu'il n'y ait plus de crises?

A ces questions, le marxisme répond dans les termes suivants : les crises étant nées avec le capitalisme lui-même, les crises étant engendrées par le développement du système capitaliste, le seul moyen d'en finir pour toujours avec les crises, c'est la suppression du système capitaliste lui-même, l'instauration d'une économie socialiste.

D'autres, par contre, prétendent qu'il est possible d'en finir pour toujours avec les crises par d'autres moyens, sans supprimer le capitalisme qui engendre les crises, sans instaurer le socialisme.

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1. Friedrich Engels : *Socialisme utopique et socialisme scientifique*, p. 68, au Bureau d'Editions, 1936.

Reminds me of Abolish/Defund the police

LES COURS DE L'UNIVERSITÉ OUVRIÈRE DE PARIS

LES GRANDS PROBLÈMES
DE LA POLITIQUE CONTEMPORAINE

par

ETIENNE FAJON

DÉPUTÉ DE LA SEINE

4.

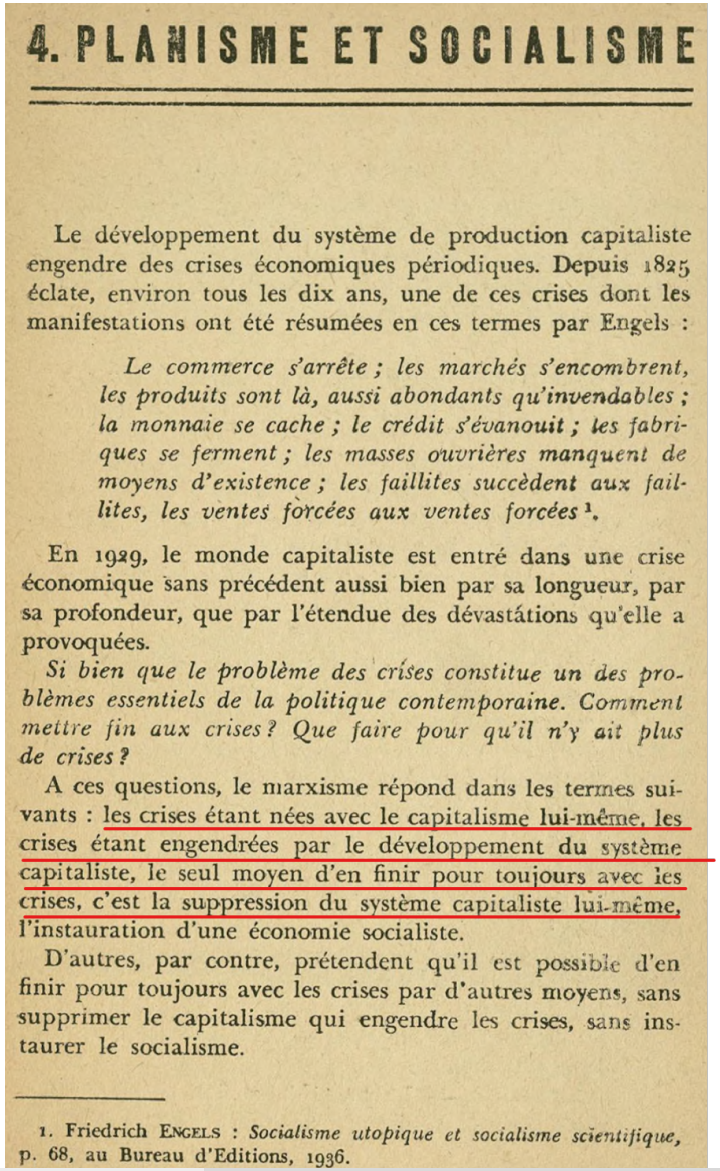
PLANISME

ET

SOCIALISME

1938

BUREAU D'EDITIONS, PARIS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1. Friedrich ENGELS : *Socialisme utopique et socialisme scientifique*, p. 68, au Bureau d'Editions, 1936.

Source of pictures

The book above is (Fajon 1938). The entire collection below can be found here.

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- He traveled back to the U.S., and then back to Berkeley to finish his degree.
- Again, he was a part of the Communist party and conducted some official businesses on behalf of the organization

(Professor May Disowns Communist Son; 'Only Honorable Course,' Says Californian 1940)

The New York Times

FRIDAY, SEPTEMBER 27, 1940

Professor May Disowns Communist Son; 'Only Honorable Course,' Says Californian

Special to THE NEW YORK TIMES.

BERKELEY, Calif., Sept. 26—Samuel Chester May, veteran University of California professor and a leader in State affairs, "disowned and disinherited" his son, Kenneth, today because he is an avowed Communist.

The youth, who is a teaching assistant at the university, is campaign manager for the Communist party in Alameda County. He made his views public last night at a meeting of the Berkeley Board of Education, when he appeared to speak in behalf of a petition for permitting the use of Berkeley school buildings for Communist meetings.

"I have been dreading this break for weeks," said Professor May. "I have been expecting it for a long time. The break came privately some time ago, but now it must come publicly.

"Every one who knows me knows that my views are contrary to those of Kenneth. So decided are my views that I have completely disowned and disinherited Kenneth."

In addition to his post as head of the university's Bureau of Public Administration Professor May is vice chairman of the State Defense Council. He went to Sacramento today and after a conference with Governor Olson explained his attitude toward his son and asserted he had taken the only honorable course in repudiating him "for his espousal of the cause of communism."

"It is just one of those things that may happen to any father," he said. "Any one who has children can understand.

"For twenty years I've been fighting communism. I have students scattered all over this country who know how I stand on radicalism and communism, so I don't believe any one will question my position.

"When I became convinced my son had become an irreconcilable Communist I took the only honorable course consistent with my personal views and the position I hold as an executive of the defense council."

Kenneth May was graduated from the University of California in 1936 after a brilliant campus record. He was a Phi Beta Kappa student and a member of the Golden Bear Honor Society and the Student Judicial Council. In competition with students throughout the United States he won a scholarship under the Institute of World Affairs and studied in Europe for two and a half years, for five months in Russia.

Following his return he was made a teaching assistant in mathematics at the university. His wife, the former Miss Ruth McGovney, daughter of Professor Dudley O. McGovney of the university, is a teacher in Oakland High School.

After the board of education meeting he said:

"Some one had to take the action I did, and what happens cannot be helped."

What a guy

- In response, May said that he believes that "the Communist party is the greatest force for good in the world and that only through Socialism can the people of the United States solve the problem of unemployment, poverty, oppression and war"

His full statement

fists. tion manager.

Statement Issued

Kenneth May Defends Membership in Party

Kenneth May, in a statement defending his membership in the Communist party, last night made no specific comment on the action of his father, Professor Samuel Chester May, in disowning him for his beliefs.

He took occasion to deny that his wife, who is the former Ruth McGovney, daughter of Professor Dudley O. McGovney of the university and a teacher in Oakland high schools, is a Communist.

May said:

"I first joined the Communist party as an undergraduate at the University of California because I found by actual experience in student activities that the Communists were consistent and uncompromising fighters for the interests of the students and against reaction within and without the university.

"I became convinced that the Communists were correct in their analysis of the university administration and the social system of which the university is a part.

"As a member of the Communist party I have seen how the party works everywhere to preserve and extend democracy, protect the living standards of the people, build the trade unions and keep this country out of war.

"During a stay of two years in Europe I observed the Communist parties in France and England in action and visited a number of other countries, including the Soviet Union. All these experiences convinced me that the Communist party is the greatest force for good in the world and that only through Socialism can the people of the United States solve the problem of unemployment, poverty, oppression and war.

"As a patriotic American I have no other choice but to work for what I consider to be the best interests of my country and the world."

Figure: (K. May 1940)

War with U.S.S.R. June 22, 1941

"All the News That's Fit to Print."

NEWS INDEX, PAGE 35, THIS SECTION

The New York Times.

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LATE CITY EDITION
Partly cloudy and continued warm today and tomorrow.
Temperatures Yesterday—Max., 91; Min., 75

Section 1

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HITLER BEGINS WAR ON RUSSIA, WITH ARMIES ON MARCH FROM ARCTIC TO THE BLACK SEA; DAMASCUS FALLS; U.S. OUSTS ROME CONSULS

MUST GO BY JULY 15

Ban on Italians Like Order to German Representatives

U. S. DENIES SPYING

Envoys Told to Protest Axis Charges—Nazis Get 'Moor' Text

By BERTRAM D. HULEN
Special to The New York Times.

WASHINGTON, June 21—The Italian Embassy was directed by the State Department in a note published today to close all its consular offices and other agencies in this country having connections with the Italian Government by July 15. This was the reply to the Italian demand for the closing of all American Consulates in Italy.

At the same time Sumner Welles, Under-Secretary of State, announced that he had sent to Dr. Hans Thomsen, the German Chargé d'Affaires, the text of President Roosevelt's message to Congress yesterday denouncing the sinking of the American freighter Robin Moor in the South Atlantic on May 21.

This message, which accused Germany of being an international outlaw, engaging in piracy and attempting to intimidate the United States by the sinking and to drive American commerce from the seas, contained notice that this country would not yield before such measures and stated that compensation would be sought for the sinking.

It was transmitted "for the information" of the German Government, but constituted in effect a note of protest. A further com-

Hope Dims for Submarine; Diver Balked at 370 Feet

Knox Believes All 33 Are Dead on the O-9 and Expects Rites at Scene for Navy 'Heroes'—Pressure Halts Descent

By RUSSELL PORTER
Special to The New York Times.

PORTSMOUTH, N. H., June 21—As hope faded rapidly for the crew of the Submarine O-9, which failed to rise after submerging yesterday morning twenty-four miles east of this city, it became known tonight that the Navy might be unable to complete its salvage operations, and might be compelled to leave the bodies entombed where they lie—440 feet below the surface of the Atlantic.

This theory was based upon the assumption that the two officers and thirty-one men must already be given up as lost, but that assumption has become stronger with every new development since the submarine was reported missing.

Last night cork insulation from the interior of the hull was picked up, showing that at least part of the submarine had collapsed, and early today, after fourteen hours of dragging, grapnels located an object believed to be the sunken craft. Since then no signals from the O-9 have been received on the sensitive sound-detection devices on the salvage ships in response to their repeated messages.

The view that the O-9's fate was sealed was strengthened this afternoon when reporters and photographers, visiting the scene in a Navy press boat, saw one of the Navy's most experienced divers fall in an attempt to reach the O-9 after descending 370 feet, or within seventy feet of where the Navy believes it has located the submarine with grapnel lines.

The diver, George Crocker, 30 years old, of Seattle, asked to be hauled up when he became convinced that he was not getting enough air pressure from his life lines of helium-oxygen mixture to overcome the increasing sea pressure as he went lower and lower.

A message from the Falcon said: "Diver descended 370 feet. Had difficulty in breathing. Brought to surface. Will continue attempts by varying diving techniques."

On the salvage ship the dive was called "the most dangerous in submarine history." It was pointed out that no one had ever made a successful "working" dive at 440 feet and that any diver who went down so far, where he would have to grope his way in complete darkness under terrific sea pressure, 195.8 pounds to the square inch, could do so only at extreme risk to his life.

Colonel Frank Knox, Secretary of the Navy, returning tonight on

Continued on Page Thirty

ARMY ASKS GUARD BE KEPT IN SERVICE

Recommends Congress Act to Hold State Troops, Reserve

NAVY MAY REPLACE SHIPYARD STRIKERS

Weighs Putting Own Machinists to Work to End Long Tie-Up

R. A. F. BLASTS FOE

Bags 26 Nazi Planes in Record Day Raids on Invasion Coast

GERMANY IS BOMBED

British on 11th Straight Night Offensive Into Western Reich

Special Cable to The New York Times.

LONDON, Sunday, June 22—Twenty-six Nazi fighter planes were destroyed in daylight yesterday by Royal Air Force fliers on their fifth straight day of raiding the Germans' invasion coast and air bases in Northern France.

Twice before dark, waves of R. A. F. warcraft—reportedly numbering at least 150 planes each—swept over the Channel in offensive operations.

Bombers attacked the Nazi's airdromes on each occasion while strong forces of fighters blasted the way for the big planes through formations of German defense fighters. While the major raids were going on, other strong R. A. F. units patrolled over the French coast and battled Messerschmitts.

Attack Goes On; Big Bombs Used

Last night and early this morning the R. A. F. was still attacking the invasion coast, using some of the latest type of high-powered bombs.

Explosions rolled across the Channel like peals of thunder, shaking the ground and rocking buildings for miles along the Kentish coast, observers there reported.

A nigh: curtain of fog hung over the Strait of Dover and little could be seen of the raids. The latest British attacks were apparently being made in the Boulogne area,

SYRIAN CITY TAKEN

French Withdraw After a Hard Fight—British Closer to Beirut

TADMUR PUSH IS ON

Allied Planes Harassing Vichy Troops, Whose Defense Falters

By C. L. SULZBERGER
By Telephone to The New York Times.

ANKARA, Turkey, June 21—French troops evacuated the city of Damascus today after a persistent bombardment by British artillery and withdrew to new positions outside the Syrian capital, according to official advices from Beirut. Early in the afternoon it was learned that the Allied vanguard was already beginning to enter the city. This evening the British reported complete occupation.

The Damascus airport at Mezze has been taken by Indian detachments of the Allied forces and one of the key points east of Damascus has been surrounded by Druz tribesmen fighting on the side of the British.

The Beirut radio announced tonight that a British motorized column pushing westward from Iraq was now heading toward Tadmur. The British column, it was said, has been bombed constantly by the French Air Force, which has just been reorganized and reinforced by French squadrons coming from North Africa. Some German planes also were said to have arrived in Syria.

Advance in High Gear

It is clear that the Allied advance is beginning to move into high gear. Unconfirmed reports that the

WHERE GERMAN ARMIES MARCH ON RUSSIA

Shown on the map is the western frontier of the Soviet Union, a battle line of more than 2,000 miles. Berlin indicated an attack from Norway to Rumania.

The Hitler Proclamation

The text of Adolf Hitler's proclamation, as recorded here by Columbia Broadcasting System, follows:

It was a difficult step for me to send my Minister to Moscow in order to attend to work against the policy of encirclement of Britain.

I hoped that at last it would be possible to put away tension.

Germany never intended to occupy Lithuania. The defeat of Poland induced me to again address a peace offer to the Allies. This was declined because Britain was still hoping to bring about European coalition

BAD FAITH CHARGED

Goebbels Reads Attack on Soviet—Ribbentrop Announces War

BALTIC MADE ISSUE

Finns and Rumanians Are Called Allies in Plan of Assault

Statement by von Ribbentrop is printed on Page 6.

By C. BROOKS PETERS
By Telephone to The New York Times.

BERLIN, Sunday, June 22—As dawn broke over Europe today the legions of National Socialist Germany began their long-rumored invasion of Communist Soviet Russia. The non-aggression and amity pact between the two countries, signed in August, 1939, forgotten, the German attack began along a tremendous front, extending from the Arctic regions to the Black Sea. Marching with the forces of Germany are also the troops of Finland and Rumania.

Adolf Hitler, in a proclamation to the German people read over a national hook-up by Propaganda Minister Dr. Joseph Goebbels at 5:30 this morning, termed the military action begun this morning the largest in the history of the world. It was necessary, he added, because in spite of his unceasing efforts to preserve peace in this area it had definitely been proved that Russia was in a coalition with England to ruin Germany by prolonging the war.

Saw Stalemate in West

Herr Hitler, in his proclamation

Bombing of Pearl Harbor Dec 7, 1941

"All the News That's Fit to Print."

The New York Times.

LATE CITY EDITION
Increasing cloudiness with rising temperature today. Tomorrow cloudy, somewhat colder.
Temperatures Yesterday—Max., 34; Min., 25

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JAPAN WARS ON U. S. AND BRITAIN; MAKES SUDDEN ATTACK ON HAWAII; HEAVY FIGHTING AT SEA REPORTED

CONGRESS DECIDED

Roosevelt Will Address It Today and Find It Ready to Vote War

CONFERENCE IS HELD

Legislative Leaders and Cabinet in Sober White House Talk

By C. P. TRUSSELL
Special to The New York Times.

WASHINGTON, Dec. 7—President Roosevelt will address a joint session of Congress tomorrow and will find the membership in a mood to vote any steps he asks in connection with the developments in the Pacific.

The President will appear personally at 12:30 P. M. Whether he would call for a flat declaration of

TOKYO ACTS FIRST

Declaration Follows Air and Sea Attacks on U. S. and Britain

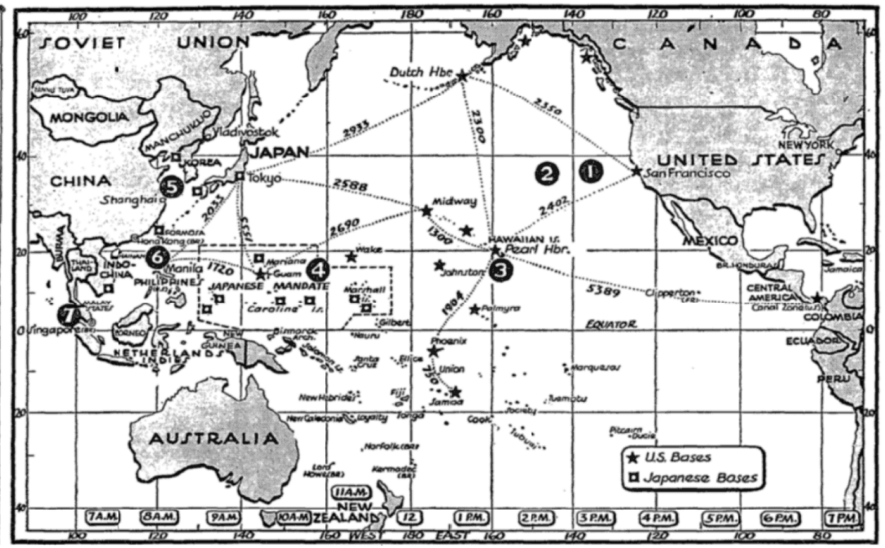
TOGO CALLS ENVOYS

After Fighting Is On, Grew Gets Japan's Reply to Hull Note of Nov. 26

By The Associated Press.

TOKYO, Monday, Dec. 8—Japan went to war against the United States and Britain today with air and sea attacks against Hawaii, followed by a formal declaration of hostilities.

Japanese Imperial headquarters announced at 6 A. M. [4 P. M. Sunday, Eastern standard time] that a state of war existed among these nations in the Western Pa-



PACIFIC OCEAN: THEATRE OF WAR INVOLVING UNITED STATES AND ITS ALLIES

GUAM BOMBED; ARMY SHIP IS SUNK

U. S. Fliers Head North From Manila—Battleship Oklahoma Set Afire by Torpedo Planes at Honolulu

104 SOLDIERS KILLED AT FIELD IN HAWAII

President Fears 'Very Heavy Losses' on Oahu—Churchill Notifies Japan That a State of War Exists

By FRANK L. KLUCKHOHN
Special to The New York Times.

WASHINGTON, Monday, Dec. 8—Sudden and unexpected attacks on Pearl Harbor, Honolulu, and other United States possessions in the Pacific early yesterday by the Japanese air force and navy plunged the United States and Japan into active war.

The initial attack in Hawaii, apparently launched by torpedo-carrying bombers and submarines, caused widespread damage and

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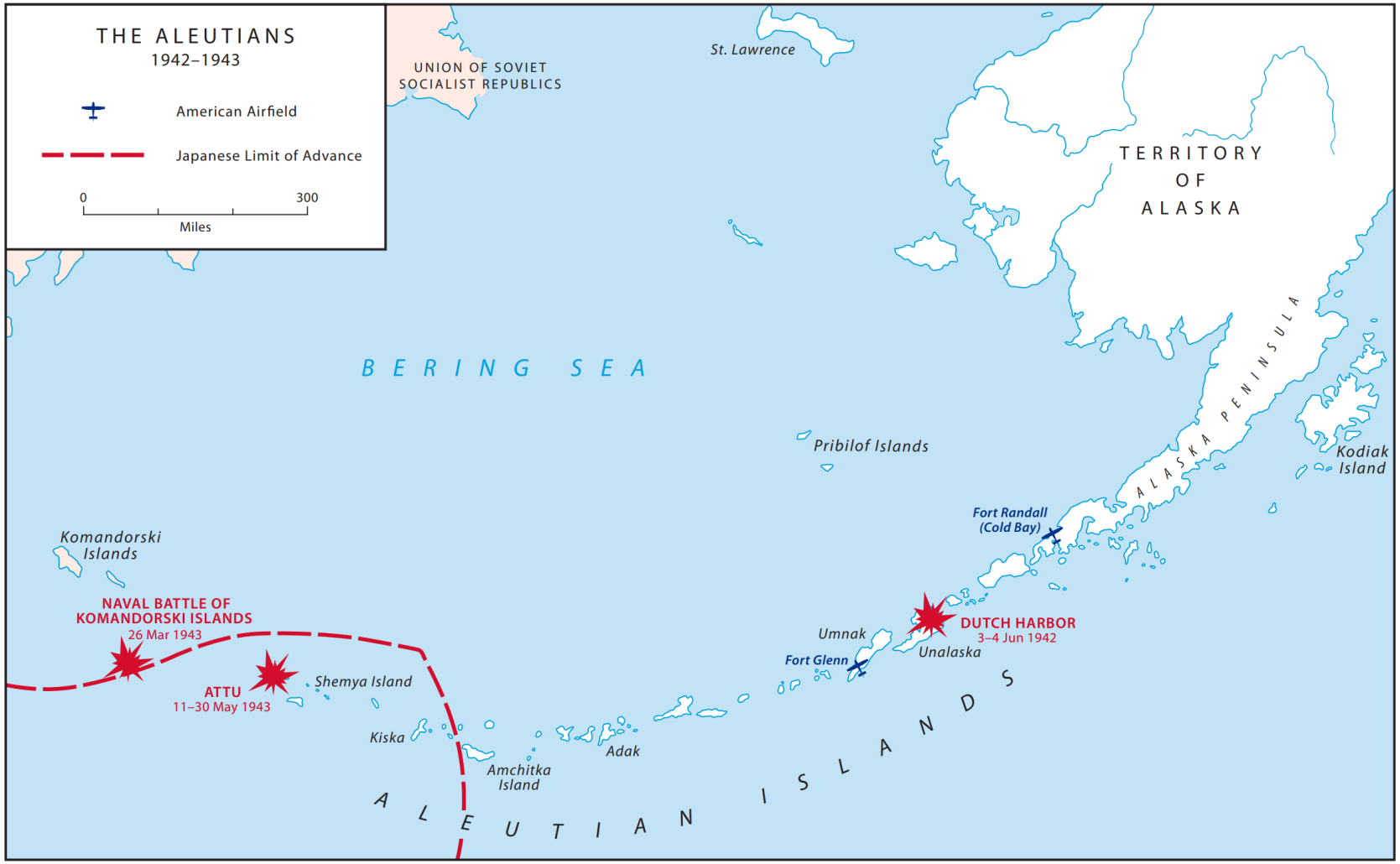
- He finally got into the army in November 1942, joining the 87th Mountain Infantry.
- His first deployment is at Kiska.

Aleutian Campaign



Source: (MacGarrigle 2019)

Sgt. May in the News

Mediterranean Rome Stars And Stripes Newspaper Archives March 3, 1945 Page 1

THE STARS AND STRIPES
MEDITERRANEAN
Vol. 2, No. 65, Saturday, March 3, 1945 Printed In Italy TWO LIRE

U. S. 9th Army Reaches Rhine

Alexander Meets With Tolbukhin; 3-Way Assault Plan Coordinated

Historic Military Events Hinted By Conference Of Two Marshals

By Sgt. STAN SWINTON
Staff Correspondent

ADVANCED ALLIED FORCE HEADQUARTERS, March 2—Field Marshal Sir Harold R. L. G. Alexander has moved to coordinate the three-pointed assault upon Ger-

HANDS ACROSS THE ADRIATIC

Yanks Capture Krefeld, Enter Neuss, Shell Ruhr

SHAEF, March 2—The watch on the Rhine was taken over by the U. S. 9th Army today.

Krefeld, great German industrial center of 165,000 inhabitants, three miles west of the Rhine and 12 miles northwest of Dusseldorf, fell tonight while U. S. 9th Army troops smashed through the crumbling enemy defenses all the way to the Rhine River somewhere in the vicinity of Neuss.

Neuss, across the river from Dusseldorf, was unofficially reported tonight to have been completely occupied.

A 9th Army spokesman announced that the German withdrawal west of the Rhine had actually turned into a "rout," while the Associated Press, in a dispatch from Field Marshal Bernard L. Montgomery's headquarters, stated tonight that "Germany's

Sgt. May in the News

Sgt. Ken May's Politics Were Stumbling Block

By. Sgt. JACK FOISIE
Staff Correspondent

What makes his Army career so interesting is that back in the days before Pearl Harbor, when Stalin and the Red Army were not in such high repute, May was a bona fide member of the Communist party in the state of California. His beliefs cost him his job as teaching assistant in mathematics at the University of California, with his dismissal being instigated by his father, Samuel May, professor of public administration at the same school.

The whole dramatic, unhappy episode made May a local cause celebre. The young educator—his-

May's application for various officer candidate schools appear to have ended up in the same blind alley. No one questioned his technical or educational qualifications, his appearance before the boards was always well received, and yet the papers came back unapproved or didn't come back at all. Once

May served as an assistant squad leader throughout the brief, Japless Kiska campaign, with the rank of Pfc. In fact it was on the direct recommendation of a three-star general that he made his stripes. It happened last November when Lt. Gen. Ben Lear visited Company A's Information and Education display. He asked May in typical general to soldier tone where he thought MacArthur would land in the Philippines.

May launched into a canny discussion of Pacific strategy in which the general soon joined. After some minutes the one-stripe and the three-star agreed on how the campaign should be won. General Lear went away highly pleased, leaving behind a recommendation that Pfc. May was deserving of at least corporal stripes.

REASON TO WONDER

"I wonder if he would have recommended me if he had known my background" May ruefully mused.

One specific act of courage attributed to May in the Mount Belvedere fight is that he dashed into a minefield to rescue a man injured in a knocked out bulldozer. May was still smiling as he debunked the feat:

"I was pretty sure the mines were all tellers. (anti-tank explosives which require much pressure to set them off) and so it wasn't as heroic as it looked."

As if for further explanation he added:

"I guess I don't have to tell you how I feel about this war."

What makes this review of May's Army career timely is that the War Department has just relaxed restrictions against Communist party members holding Army commissions.

"Yes," said May scratching his beard, "I would like to be an officer if the opportunity comes along."

Source: (Foisie 1945)

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- He founded *Historia Mathematica* in 1974, a journal still in circulation today.

Summary

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    3. May: the only resolute, neutral, anonymous, and positively responsive two-candidate SCF is simple majority rule

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3. Kenneth May—the patriot, Communist, mathematician, and historian.

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